DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY
Faculty of Science & Technology
Bangladesh University of Professionals (BUP)
Course Title: **Digital Communication (ICE 3105)** (3rd Year, 1st Semester, January-June, 2026)
Class Test-04

Time: **20 minutes** Full Mark: **10**

1. Define $h(Y|X)$. 1

2. Explain the properties of AWGN Noise. 3

3. Explain how the coherent receiver recovers a signal in a receiver. Note that providing an example of mathematical derivation is mandatory. 4

4. What are the disadvantages of a non-coherent receiver over a coherent receiver? 2